

~minutesheet~

Proof by Induction

Example

Claim: If $\text{succ}(n) \text{ nat}$ then $n \text{ nat}$

Proof:

Proof by induction on $\text{succ}(n) \text{ nat}$

[Case : zero]

GOAL = If $\text{succ}(\text{zero}) \text{ nat}$ then zero nat

First we assume the premise

$P = \text{succ}(\text{zero}) \text{ nat}$

GOAL' = zero nat

$$\frac{}{\text{zero nat}} Z \quad \frac{n \text{ nat}}{\text{succ } n \text{ nat}} S$$

zero nat by the Z rule.

$$\frac{}{\text{zero nat}} Z$$

□_{zero}

[Case : succ]

GOAL = if $\text{succ}(\text{succ } n) \text{ nat}$
then $\text{succ } n \text{ nat}$

$P = \text{succ}(\text{succ } n) \text{ nat}$

GOAL' = $\text{succ } n \text{ nat}$

IH = if $\text{succ } n \text{ nat}$
then $n \text{ nat}$

By P we have $\text{succ}(\text{succ } n) \text{ nat}$, which
must have had the following derivation

$$\frac{\frac{\frac{}{\text{succ } n \text{ nat}} Z}{\text{succ } n \text{ nat}} S}{\text{succ}(\text{succ } n) \text{ nat}} S$$

The derivation of P contains a
derivation of our goal: ①

D_{succ}

□

Example

premise

Claim: If n even then either
 $n = \text{zero}$, or
 $n = \text{succ } x$, where x odd

Proof: (CLAIM-E) $\rightarrow$ conclusion

Proof by simultaneous induction on n even.

Complementary claim for odd numbers:

If n odd then $n = \text{succ}(x)$ where x even. (CLAIM-O)

[Case: zero]

GOAL = If $\overset{n}{\text{zero}}$ even
then either
 $n = \text{zero}$, or
 $n = \text{succ}(x)$, where x odd

This is immediate as $n = \text{zero}$ $\square_{\text{zero}}$

[Case: EVEN]

GOAL = if $\text{succ } n$ even
then either
 $\text{succ } n = \text{zero}$, or
 $\text{succ } n = \text{succ } x$ where
 x odd

$P = \text{succ } n$ even

GOAL' = $\text{succ } n = \text{succ } x$ where x odd
 $= n$ odd

[IH = CLAIM-O for n
 $=$ if n odd
then $n = \text{succ } x$, where x even]

We can prove our goal by looking
at P's derivation:

: ①
 $\frac{n \text{ odd}}{\text{succ } n \text{ even}} \text{ EVEN}$

① is a derivation of our GOAL'

$\square_{\text{EVEN}}$

[Case: ODD]

GOAL = CLAIM-O @ $\text{succ } n$
 $=$ if $\text{succ } n$ odd
then $\text{succ } n = \text{succ } x$ where
 x even

$P = \text{succ } n$ odd

GOAL' = $\text{succ } n = \text{succ } x$, where x even

We can achieve a derivation of
our goal.

$\frac{n \text{ even}}{P = \text{succ } n \text{ odd}} \text{ ODD}$
 $\square_{\text{ODD}}$
 $\square$